

# Where Does $p \geq \frac{1}{2}$ Come From?

## The Setup

Let me show you where the  $p \geq \frac{1}{2}$  condition actually comes from by deriving the FOC.

## The Problem

You have initial wealth  $w_0$ . You choose how much  $x$  to invest in a risky asset:

- With probability  $p$ : risky asset succeeds, you get back  $Rx$  (where  $R > 1$  is the return)
- With probability  $(1 - p)$ : risky asset fails, you get back 0
- The remaining  $(w_0 - x)$  stays safe

Your final wealth is:

- **Good state (prob  $p$ ):**  $w_G = w_0 - x + Rx = w_0 + (R - 1)x$
- **Bad state (prob  $1 - p$ ):**  $w_B = w_0 - x$

Your utility: CRRA with  $u(w) = \frac{w^{1-\sigma}}{1-\sigma}$

## Setting Up Expected Utility

$$\begin{aligned} EU(x) &= p \cdot u(w_G) + (1 - p) \cdot u(w_B) \\ &= p \cdot \frac{[w_0 + (R - 1)x]^{1-\sigma}}{1 - \sigma} + (1 - p) \cdot \frac{[w_0 - x]^{1-\sigma}}{1 - \sigma} \end{aligned}$$

## Taking the First-Order Condition

Take derivative with respect to  $x$  and set equal to zero:

$$\begin{aligned} \frac{dEU}{dx} &= p \cdot \frac{[w_0 + (R - 1)x]^{-\sigma}}{1 - \sigma} \cdot (R - 1) \\ &\quad + (1 - p) \cdot \frac{[w_0 - x]^{-\sigma}}{1 - \sigma} \cdot (-1) = 0 \end{aligned}$$

Cancel out  $\frac{1}{1-\sigma}$  (assuming  $\sigma \neq 1$ ):

$$p(R - 1)[w_0 + (R - 1)x]^{-\sigma} = (1 - p)[w_0 - x]^{-\sigma}$$

## Rearranging

$$\frac{p(R-1)}{1-p} = \frac{[w_0 - x]^{-\sigma}}{[w_0 + (R-1)x]^{-\sigma}}$$
$$\frac{p(R-1)}{1-p} = \left( \frac{w_0 + (R-1)x}{w_0 - x} \right)^{\sigma}$$

Raise both sides to power  $\frac{1}{\sigma}$ :

$$\left[ \frac{p(R-1)}{1-p} \right]^{1/\sigma} = \frac{w_0 + (R-1)x}{w_0 - x}$$

## Solving for $x$

Let  $K = \left[ \frac{p(R-1)}{1-p} \right]^{1/\sigma}$  for simplicity.

$$K(w_0 - x) = w_0 + (R - 1)x$$

$$Kw_0 - Kx = w_0 + (R - 1)x$$

$$Kw_0 - w_0 = Kx + (R - 1)x$$

$$w_0(K - 1) = x[K + (R - 1)]$$

**The Solution:**

$$x^* = \frac{w_0(K - 1)}{K + (R - 1)}$$

where  $K = \left[ \frac{p(R-1)}{1-p} \right]^{1/\sigma}$

## When is $x^* \geq 0$ ?

For  $x^* \geq 0$ , we need:

$$\frac{w_0(K - 1)}{K + (R - 1)} \geq 0$$

Since  $w_0 > 0$  and  $K + (R - 1) > 0$  always (both positive), we need:

$$K - 1 \geq 0 \quad \Rightarrow \quad K \geq 1$$

That is:

$$\left[ \frac{p(R - 1)}{1 - p} \right]^{1/\sigma} \geq 1$$

Raise both sides to power  $\sigma$  (assuming  $\sigma > 0$ ):

$$\frac{p(R - 1)}{1 - p} \geq 1$$

$$p(R - 1) \geq 1 - p$$

$$p(R - 1) + p \geq 1$$

$$pR \geq 1$$

**General Condition:**

$$p \geq \frac{1}{R}$$

For  $x^* \geq 0$ , you need the probability of success to be at least  $\frac{1}{R}$ .

## The Special Case: $R = 2$

In your specific problem, if the risky asset **doubles your money** (pays back \$2 for every \$1 invested), then  $R = 2$ .

Substituting:

$$p \geq \frac{1}{2}$$

That's where the  $p \geq \frac{1}{2}$  comes from!

## Intuition

### General Intuition:

For the risky asset to be worth investing in at all, its **expected gross return** must be at least 1:

$$pR + (1 - p) \cdot 0 \geq 1$$

This gives:  $pR \geq 1$ , or  $p \geq \frac{1}{R}$

**In words:** "The probability of success times the payoff must be at least 1 for me to break even in expectation."

## Examples

### Example 1: $R = 2$ (doubles your money)

- Need  $p \geq \frac{1}{2}$  (succeeds at least 50% of the time)
- Expected return:  $2p \geq 1$

### Example 2: $R = 4$ (quadruples your money)

- Need  $p \geq \frac{1}{4}$  (succeeds at least 25% of the time)
- Even though success rate is lower, the bigger payoff ( $R = 4$ ) compensates

### Example 3: $R = 1.5$ (50% return)

- Need  $p \geq \frac{1}{1.5} = \frac{2}{3} \approx 0.67$  (succeeds at least 67% of the time)
- Smaller payoff means you need higher success probability

## What if $p < \frac{1}{R}$ ?

If  $p < \frac{1}{R}$ , then the FOC would give  $x^* < 0$ .

**What does negative  $x$  mean?**

- Mathematically: you want to "invest negative amount" = short sell the risky asset
- In reality: you can't do this (constraint  $x \geq 0$ )

So when  $p < \frac{1}{R}$ , the optimal choice given the constraint is:

$$x^* = 0 \quad (\text{corner solution})$$

**Intuition:** The risky asset is such a bad bet (low probability of success relative to the payoff) that you shouldn't invest at all.

## Summary

**The Complete Picture:**

**General condition:**  $p \geq \frac{1}{R}$  for  $x^* \geq 0$

**Your specific problem:** If  $R = 2$ , then  $p \geq \frac{1}{2}$

**Decision rule:**

- If  $p \geq \frac{1}{R}$ : Use the FOC to get interior solution  $x^* > 0$
- If  $p < \frac{1}{R}$ : Corner solution, set  $x^* = 0$  (don't invest)

**Why?** When  $p < \frac{1}{R}$ , the expected gross return  $pR < 1$ , meaning you expect to lose money on average. A risk-averse person definitely won't invest!

## Check Your Understanding

**Q: Why do we need  $p \geq \frac{1}{R}$  and not just  $pR > 0$ ?**

A: Because we also keep  $(w_0 - x)$  safe. The risky asset needs to beat the **outside option** of keeping money safe (which has return 1). So we need:

$$\text{Expected return of risky asset} = pR \geq 1$$

**Q: Does this mean if  $p \geq \frac{1}{R}$ , everyone invests the same amount?**

A: No! The exact amount  $x^*$  depends on:

- $p$  (higher  $p \Rightarrow$  invest more)
- $\sigma$  (higher risk aversion  $\Rightarrow$  invest less)
- $R$  (higher return  $\Rightarrow$  invest more)
- $w_0$  (more initial wealth  $\Rightarrow$  invest more in absolute terms)